

Section 4.3 (cont.)

I. Curve Sketching $y = f(x)$

- a. Identify the domain
- b. Identify any intercepts
- c. Identify the critical numbers: $f'(x)$ vanishes
- d. Identify the intervals of monotonicity:
 1. Intervals where $f'(x) > 0 \Leftrightarrow$ Intervals where f is strictly increasing
 2. Intervals where $f'(x) < 0 \Leftrightarrow$ Intervals where f is strictly decreasing
- e. Classify each critical point as a relative maximum, relative minimum or neither
- f. Identify the second-order critical numbers: $f''(x)$ vanishes
- g. Identify the intervals of concavity
 1. Intervals where $f''(x) > 0 \Leftrightarrow$ Intervals where f is concave up
 2. Intervals where $f''(x) < 0 \Leftrightarrow$ Intervals where f is concave down
- h. Identify any inflection points
- i. Construct a table with a few specific values of the function
- j. Sketch the graph of $y = f(x)$ incorporating all of the above information